

HTML5 Semantic Tags & Form Validation

1. Why Semantic Tags Matter

Definition: Semantic tags are HTML elements whose name describes the meaning of the content inside them, not just how it looks. `<div>` and `<span>` are non-semantic — they say nothing about what's inside; `<header>`, `<nav>`, `<main>` etc. tell the browser, search engines, and screen readers exactly what role that section plays on the page.

Example: Before HTML5, developers wrote `<div id='header'>`, `<div id='nav'>`, `<div id='footer'>` — this worked visually but gave no real meaning. HTML5 semantic tags replace those `div/id` patterns with tags that carry meaning by themselves.

Recap / Extra: Benefits: better SEO (search engines understand page structure), better accessibility (screen readers can jump straight to `<nav>` or `<main>`), and cleaner, more readable code for other developers.

2. header, nav, main, footer

Tag	Purpose
<header>	Introductory content for a page or a section — usually a logo, site title, or heading group
<nav>	A block of navigation links — the main menu, breadcrumbs, or a table of contents
<main>	The one, unique, central content of the page — excludes repeated header/nav/footer. Only one per page.
<footer>	Closing content — copyright, contact links, site map — usually repeated at the bottom of every page

```
<header>
  <h1>GLOBAL Group</h1>
</header>
<nav>
  <a href="#">Home</a> | <a href="#">About</a> | <a href="#">Contact</a>
</nav>
<main>
  ...page content...
</main>
<footer>
  <p>&copy; 2026 GLOBAL Group</p>
</footer>
```

Recap / Extra: A page can have more than one <header>, <nav>, or <footer> (e.g. one per <article>), but only one <main>.

3. section vs article vs aside

Tag	Purpose	Rule of thumb
<section>	A thematic grouping of related content, usually with its own heading	Use when content belongs together under one topic
<article>	Self-contained content that would make sense on its own, out of context	Use if you could copy-paste it onto another site and it would still make sense (a blog post, a product card, a news story)
<aside>	Content indirectly related to the main content — sidebars, pull-quotes, ads	Use for anything the reader could skip without losing the main point

```
<section>
  <h2>Latest Articles</h2>
  <article>
    <h3>Understanding Semantic HTML</h3>
    <p>Article content...</p>
  </article>
  <article>
    <h3>Form Validation Basics</h3>
    <p>Article content...</p>
  </article>
</section>
<aside>
  <p>Related: HTML vs XHTML</p>
</aside>
```

Example: <section> and <article> are often confused: a whole blog page can be a <section> titled 'Blog', while each individual post inside it is its own <article> because each post stands on its own.

4. Task 1 — Full Semantic Page Layout

Combining header, nav, main, section, article, aside, and footer into one complete page structure.

```
<body>
  <header>
    <h1>GLOBAL Tech Blog</h1>
  </header>

  <nav>
    <a href="#">Home</a> | <a href="#">Articles</a> | <a href="#">Contact</a>
  </nav>

  <main>
    <section>
      <h2>Recent Articles</h2>
      <article>
        <h3>Why Semantic HTML Matters</h3>
        <p>Semantic tags improve accessibility and SEO...</p>
      </article>
    </section>

    <aside>
      <h3>Related Links</h3>
      <p>HTML5 Form Validation Guide</p>
    </aside>
  </main>

  <footer>
    <p>&copy; 2026 GLOBAL Group. All rights reserved.</p>
  </footer>
</body>
```

5. Other Semantic Tags Worth Knowing

These weren't explicitly asked for in the task list but are commonly grouped with HTML5 semantic tags and often come up in the same viva.

Tag	Purpose
<figure> / <figcaption>	Groups an image (or diagram/code snippet) with its caption as one unit
<details> / <summary>	Creates a native expand/collapse widget with no JavaScript needed

```
<figure>
  
  <figcaption>Fig 1: A typical semantic page layout</figcaption>
</figure>

<details>
  <summary>What is semantic HTML?</summary>
  <p>HTML that describes meaning, not just appearance.</p>
</details>
```

6. Task 2 — Contact Form: label, input, textarea, button

Element	Purpose
<label>	Describes a control and links to it via for/id, improving accessibility
<input>	A single-line control; type decides text, email, tel, etc.
<textarea>	A multi-line control, for a longer message
<button>	A clickable control — type='submit' sends the form

```
<form>
  <label for="cName">Name</label>
  <input type="text" id="cName" name="cName">

  <label for="cEmail">Email</label>
  <input type="email" id="cEmail" name="cEmail">

  <label for="cMessage">Message</label>
  <textarea id="cMessage" name="cMessage" rows="4"></textarea>

  <button type="submit">Send Message</button>
</form>
```

7. Task 3 — required Attribute (Name, Email, Phone Number)

required: A boolean attribute — its presence alone blocks form submission until that field has a value. The browser shows a built-in warning message if it's left empty.

```
<form>
  <label for="fName">Name</label>
  <input type="text" id="fName" name="fName" required>

  <label for="fEmail">Email</label>
  <input type="email" id="fEmail" name="fEmail" required>

  <label for="fPhone">Phone Number</label>
  <input type="tel" id="fPhone" name="fPhone" required>

  <button type="submit">Submit</button>
</form>
```

Recap / Extra: required works with almost every input type (text, email, tel, checkbox, radio, select, textarea) — for radio/checkbox groups, only one input in the group needs it for the group to be enforced.

8. Task 4 — Registration Form: type="email", type="number", pattern

Feature	What it validates
type='email'	Checks the value roughly matches an email shape (text@text.text) before allowing submission
type='number' + min/max	Restricts entry to numbers, and to a numeric range
pattern	Checks the value against a custom regular expression — for rules a type alone can't express

```
<form>
  <label for="rEmail">Email</label>
  <input type="email" id="rEmail" name="rEmail" required>

  <label for="rAge">Age (18-60)</label>
  <input type="number" id="rAge" name="rAge" min="18" max="60" required>

  <label for="rPassword">Password</label>
  <input
    type="password" id="rPassword" name="rPassword"
    pattern="(?!.*[A-Z])(?!.*[0-9]).{8,}"
    title="At least 8 characters, one uppercase letter and one number"
    required>

  <button type="submit">Register</button>
</form>
```

Example: The pattern above requires at least 8 characters, containing at least one uppercase letter and one digit. The title attribute isn't just documentation — browsers display it as the validation error tooltip when the pattern doesn't match.

9. Task 5 — Feedback Form: minlength, maxlength, min, max

Attribute	Applies to	Meaning
minlength	Text-based inputs, textarea	Minimum number of characters required
maxlength	Text-based inputs, textarea	Maximum number of characters allowed (browser blocks further typing)
min	number, range, date-type inputs	Minimum numeric/date value allowed
max	number, range, date-type inputs	Maximum numeric/date value allowed

```
<form>
  <label for="fbComment">Comments (10-200 characters)</label>
  <textarea id="fbComment" name="fbComment" minlength="10" maxlength="200"
required></textarea>

  <label for="fbRating">Rating (1-5)</label>
  <input type="number" id="fbRating" name="fbRating" min="1" max="5" required>

  <button type="submit">Submit Feedback</button>
</form>
```

Recap / Extra: minlength/maxlength count characters (for text). min/max compare values (for numbers/dates/range) — easy to mix up since they sound similar but measure different things.

10. All Validation Attributes — Summary Table

Attribute	Meaning
required	Field must have a value before the form can submit
type (email/number/tel/url)	Restricts and shape-checks the kind of value entered
pattern	Custom regular expression the value must match
minlength / maxlength	Minimum/maximum character count for text-based fields
min / max	Minimum/maximum value for number, range, and date-type fields
step	The allowed increment for number/range fields, e.g. step='5'

11. Where This Fits in the Course Workflow

Following the course progression HTML -> CSS -> Bootstrap -> Tailwind CSS, semantic tags define structure now; layout and validation styling come from CSS and the frameworks next.

This lab (semantic HTML5)	CSS (next labs)	Bootstrap / Tailwind (later labs)
header / nav / main / footer	Flexbox or Grid for layout positioning	Bootstrap: navbar, container Tailwind: flex, grid
Default browser error tooltips	:invalid / :valid pseudo-classes for custom styling	Bootstrap: is-invalid classes Tailwind: form validation patterns
Plain <button>	Custom button styling with CSS	Bootstrap: btn btn-primary Tailwind: bg-blue-600 rounded px-4

Example: Keep this lab's semantic structure and validation attributes exactly as they are — the CSS lab will style :invalid fields in red and lay out the header/nav/main/footer visually, without changing any of the HTML written here.

12. Revision Quiz

1. What is the difference between a semantic tag and a non-semantic tag like `<div>`?
 2. How many `<main>` elements can a single page have?
 3. Give one example each of content that belongs in `<section>`, `<article>`, and `<aside>`.
 4. What does the `required` attribute do, and does it need a value?
 5. How does `type='email'` help with validation compared to `type='text'`?
 6. Write a pattern that would require exactly 6 digits.
 7. What is the difference between `minlength/maxlength` and `min/max`?
 8. What does the `title` attribute do when used alongside pattern?
 9. What is the purpose of `<figcaption>`, and which tag must it be placed inside?
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